EATA: Bridging Tail-Risk Objectives and Symbolic Search for Transparent Trading

Jingtong Zhang[a], Xiaoang Yang[b], Yinqi Shi[c], Yin Tang[d,]

[a] *University of New South Wales, Sydney, 2052, Australia*
[b] *School of Cybersecurity, Jinan University, GuangZhou, 511443, China*
[c] *Jinan University – University of Birmingham Joint Institute, Jinan University, GuangZhou, 511443, China*
[d] *School of Management, Jinan University, GuangZhou, 510632, China*

Abstract

Profit-seeking yet interpretable trading remains elusive: deep neural predictors excel at accuracy but violate regulatory transparency, whereas symbolic regression yields readable formulas but optimizes proxy objectives misaligned with economic utility. We present EATA (Explainable Algorithmic Trading Agent), a neural-guided symbolic regression framework that aligns objective, hypothesis space, and training signal with trading profitability. EATA couples a three-headed policy–value–profit network (PVPNet) with Monte Carlo Tree Search so that accuracy rewards r^{acc} stabilize search while a Wasserstein-derived profit reward r^{rl} captures asymmetric tail risk. A finance-grounded, decoupled grammar with bounded module libraries encodes moving-average, momentum, RSI, volatility, and conditional operators to curb hypothesis-space inflation, while sliding-window inheritance and quantile-consensus signals enable temporal adaptation. Across representative S&P 500 constituent stocks (N=62 stocks) (2013–2018), EATA delivers annualized return 21.2%, Sharpe ratio 0.76, and maximum drawdown 36.1%, outperforming strong baselines while maintaining full interpretability; Pareto-frontier analyses confirm a favorable complexity–performance trade-off. Ablation and objective studies show that removing Wasserstein rewards, neural guidance, or grammar augmentation severely degrades performance, demonstrating that profit-aware symbolic regression can achieve interpretable trading strategies competitive with black-box counterparts.

Keywords: Algorithmic Trading, Symbolic Regression, Wasserstein Distance, Explainable AI, Optimal Transport, Risk-Sensitive Learning

1. Introduction

Algorithmic trading is in a tense zone between two competing core requirements: although modern deep learning and reinforcement learning systems have achieved the most advanced prediction accuracy [1, 2], they are not transparent inductive bias. It violates emerging regulatory and governance requirements such as MiFID II [3], and also increases the difficulty of risk supervision [4, 5, 6]. Therefore, practitioners not only hope that the trading strategy can be explained, but also hope that it can cope with the switching of the market state, especially because the financial income sequence has the characteristics of heavy tailed and non-stationary [7]. The scientific problem here is not only "how to explain a black box", but also *how to discover profit-aligned trading rules whose structure itself is the explanation.*

The existing research provides three incomplete solutions. The Black-box DL/RL models has strong original performance, but it provides almost no transparency or security guarantee. Classic symbolic regression and alpha-mining pipelines can return readable formulas [8, 9], but they usually optimize proxy goals that are disconnected from economic benefits (MSE, IC) , which will amplify the multiple-testing bias and turnover costs. Hybrid grammar-guided systems (such as AlphaCFG [10], NEMoTS [11]) constrain the search space, but still inherits the point-loss objectives and lacks mechanisms for temporal adaptation. These shortcomings prompt us to design a profit-aware, grammar-disciplined discovery agent.

Therefore, we raise the core research question: *Can we design a symbolic regression agent so that its goals, hypothetical space and training signals are aligned with trading profitability under regulatory and regime constraints?* This problem can be broken down into three sub-problems:

RQ1 **Objective alignment.** Which distributional divergence can better capture the asymmetric tail risk than point-wise losses, and how should it regularize the search process without overfitting noise?

RQ2 **Search discipline.** How to integrate neural priors, grammar constraints and inheritance mechanisms to maintain auditability and adapt to market shifts while exploring large-scale symbol spaces?

RQ3 **Empirical closure.** Considering the transaction costs and the S&P 500 universes, can such an intelligent body obtain risk-adjusted returns equivalent to

Email address: `Corresponding author: Yin Tang (Email: ytang@jnu.edu.cn).` (Yin Tang)

or better than the black-box baselines?

In order to answer these questions, we introduce three innovative points of interaction: (i) a Wasserstein-guided policy-value-profit network (PVPNet), which decouples the accuracy reward r^{acc} from the profit-aware reward r^{rl}, and injects the latter into the MCTS back-ups; (ii) a decoupling syntax based on finance, including discrete window/lag non-terminals, conditional logic and a bounded module library to alleviate the hypothesis-space inflation; (iii) a sliding-window pipeline with expression inheritance, module extraction and quantile consensus signals to achieve temporal adaptation.

Our contributions are as follows:

1. **Profit-aware symbolic regression.** We model the discovery of trading rules into a distribution matching through the 1-Wasserstein distance, and couple the profit rewards with the accuracy stabilizers to control the tail risk under heavy-tailed returns.
2. **Neural-guided grammar search with temporal adaptation.** We have integrated three-headed PVPNet, grammar constraints, module libraries and sliding window inheritance mechanisms to explore hypothesis spaces that are both expressive and auditable under regime shifts.
3. **Comprehensive empirical validation.** We have provided many years of multi-universe experiments, transaction-cost robustness analysis and ablations, which directly answered RQ1 to RQ3 and proved the competitiveness compared with the black-box baselines.

Section 2 situates EATA within evaluation protocols, symbolic discovery, and risk-sensitive learning; Section 3 elaborates the framework; Section 4 presents the evidence; Section 5 discusses limitations and future work.

2. Related Work

This section places EATA at the intersection of three areas: (i) empirical algorithmic trading evaluation, (ii) symbolic regression and neural-guided search, and (iii) distributional / risk-sensitive objectives. We organize the existing work according to the pragmatic taxonomy aligned with the research questions: **(A)** evaluation and backtesting protocols, **(B)** symbolic discovery and search algorithms, **(C)** distributional metrics and risk-sensitive learning.

2.1. Evaluation, Backtesting, and Statistical Validity in Trading

Empirical algorithmic trading research usually reports risk-adjusted performance metrics derived from classical portfolio theory [12, 13]. However, in a limited sample, the estimated value of the Sharpe ratio may have considerable uncertainty and non-Gaussian sensitivity [14, 15], and the naive benchmark selection may trigger multiple-testing and data-snooping bias. Classical econometric methods, such as reality check [16] and superior predictive ability (SPA) test [17], provide a principled benchmark for large-scale strategy comparison; phase The work further emphasizes the need to correct the selection bias and backtest overfitting [18, 19]. In the broader asset pricing literature, when the factor libraries are mined on a large scale, multiple test questions have been recorded in large numbers [20]. In addition to statistical validity, realistic transaction cost modeling is also crucial, because once many strategies are introduced into implementation frictions, their performance will drop sharply [21]. These considerations prompt us to emphasize risk-adjusted metrics and robustness checks, rather than just focusing on forecast accuracy.

A recurring trap in the trading literature is to equate the in-sample predictive performance with the out-of-sample trading profitability. This misalignment is further amplified because the evaluation target (returns) itself is noisy, heavy-tailed and regime-dependent [7]. Therefore, small modeling changes may also lead to unstable performance rankings of different assets and different periods.

These problems prompt EATA to follow two design principles: (i) to reduce implicit multiple testing by grammatically constraining the hypothesis space, and (ii) aligning learning signals with economically significant results to avoid proxy-driven improvements.

Finally, we point out that algorithmic trading is not only a modeling problem, but also a market structure problem: automated execution and algorithmic strategies may affect liquidity and price dynamics [22]. This further promotes our adoption of an assessment process that emphasizes robustness and auditability.

2.2. Interpretable Alpha Discovery and Formulaic Trading Signals

A long-standing research direction focuses on human-interpretable and formulaic trading signals and alpha factors. Technical analysis has been widely reviewed, but the evidence of its profitability is vary, and it shows significant sensitivity to market regimes and evaluation protocols [23]. The recent "alpha mining" work aims to automatically discover and combine formulaic factors on a large scale, covering from alpha discovery based on genetic programming [24] to reinforcement learning and languege-model assisted frameworks [25, 26, 27] and other methods. These methods reinforce a key methodological view: interpretability does not automatically mean economic usefulness unless the objective is found to be aligned with the utility of the trading.

The Alpha discovery system usually implies an assumption that factors that can improve proxy objective (such as prediction loss, IC or simplified RL reward) will be transformed into a profitable and stable trading rule. In practice, this assumption will be broken under at least three common constraints. (i) Turnover and costs: Weak forecast signals near zero may trigger frequent transactions and fail under the realistic friction models [21]. (ii) Tail constraints: Many practitioners operate under the drawdown limits; the factors that occasionally generate large losses may be unacceptable even if the mean returns is positive. (iii) Hypothesis-space inflation: large-scale factor libraries make error detection more likely to occur, unless the search process is explicitly regularized and the evaluation process takes into account multiple testing [16, 20].

EATA can be regarded as a "constrained alpha miner": it retains the formulaic interpretability, but binds the discovery process to a profit-aware and distribution-sensitive learning signal, while using grammar constraints to limit the degrees of freedom.

Therefore, we will find that goals (profit perception, distribution sensitivity) and hypothesis-space control (grammar constraints) are regarded as first-class components, rather than realizing details.

2.3. Black-Box Learning for Trading and the Interpretability Gap

Deep learning and reinforcement learning have been widely used in trading, including recurrent architectures for financial forecasting [1], transformers for multi-scale pattern extraction [28, 29], and the deep RL systems for sequential decision making [30, 31]. Modern chronological Transformer variants, such as Informer, Autoformer and PatchTST, have shown higher efficiency in long-serial financial forecasts; we use a vanilla Transformer as a representative baseline, because we The focus is on discovery objective rather than architectural innovation, and grammar-constrained symbolic expressions run on pre-calculated technical indicators, not directly on the original sequence. These models can achieve strong empirical performance, but their inductive biases is largely opaque, and they may be very sensitive to the choice of non-stationarity and evaluation protocol.

In addition to neural models, tree ensembles are still highly competitive in the financial prediction pipelines because of their sample efficiency and robustness to heterogeneous features. Gradient boosting such as XGBoost [32] and LightGBM [33] are widely used baseline methods in practice. However, even if the post-hoc explanation tools are applied, the integrated model may produce unstable interpretations between relevant characteristics or under regime shifts. Rule extraction methods such as inTrees [34] partially bridge this gap, but they usually explain a fixed predictor, rather than directly optimize the discovery for interpretable and deployable trading rules.

Black-box literature usually assumes that under deployment constraints, the improvement of predictive capacity will translate into an improvement in trading performance. However, in the financial field, this assumption is particularly fragile, because: (i) low signal-to-noise ratios will amplify overfitting, (ii) regime shifts destroy stationarity, (iii) decision-making must meet auditability and compliance constraints [3, 5, 6]. In addition, although the post-hoc explanation methods (LIME/SHAP) can provide local rationales, it may not meet the requirements of robustness and faithfulness, and may even be manipulated [35, 36, 37]. This prompts us to adopt intrinsically interpretable strategies, that is, the decision-making rules themselves are models.

2.4. Symbolic Regression for Time Series: Evolutionary, Grammar-Based, and Neural

Evolutionary and Grammar-Based GP. Genetic Programming (GP) is still the basic paradigm of symbolic regression [8, 38]. Grammar-based GP provides a principled way to code domain constraints and reduce invalid expressions [39]. Modern toolkits such as gplearn [40] and Operon [41] improve scalability through efficient evaluation. In the field of computational finance, evolutionary methods have been widely explored for alpha discovery and rule mining [42], but they often suffer from bloat and expensive fitness evaluations, especially when the target function This is especially true when the actual backtesting is needed.

Neural Symbolic Regression. Recent studies have introduced neural guidance to improve sample efficiency and scalability. DSR [43] uses the policy gradients to generate expressions token-by-token, and the neural-to-symbolic models based on transformers has shown promising extended behavior on symbolic tasks [44, 45, 46]. The comprehensive review emphasizes that a core challenge is still to design the target function that can reflect the downstream utility of the discovered expressions, not just the reconstruction error [47].

In addition to direct expression generation, the grammar-aware generative models provide another way to impose syntactic constraints while learning expressive priors. For example, the grammar variational autoencoders [48] shows how to use context-free grammars to ensure syntactic validity. In addition, program synthesis and library learning methods such as DreamCoder [49] learn reusable subroutines, so as to improve sample efficiency and compress hypothesis space. These ideas are conceptually consistent with EATA's module extraction: instead of learning an implicit program a priori, we explicitly maintain a bounded library of reusable sub-expressions to maintain auditability.

For time series trading, there are two practical constraints that are often ignored. **(i) Evaluation-dominated cost:** In the transaction, the expensive part is not to parse the expressions, but to evaluate them under the backtesting protocol (positions, costs and risk metrics). Therefore, the method of generating a large number of candidates without strong pruning is computationally not feasible. **(ii) Spuriousness under non-stationarity:** If there is no clear domain constraint, symbolic regression may fit a transient correlation that will disappear under the regime shifts; even if the expressions are explainable, they may be unstable.

As a structural prior, grammatical constraints in EATA are both scientific and practical: from a scientific point of view, they make domain knowledge operable and limit the hypothesis class; from a practical point of view, they reduce the scale of multiple-testing problem caused by search.

We have explicitly coded financial operators and fixed horizons to trade-off between expressiveness for robustness and auditability.

2.5. Neural-Guided Search and MCTS for Symbolic Discovery

Monte Carlo Tree Search (MCTS) [50, 51] provides a principled exploration–exploitation trade-off, and has been successfully combined with neural policy–value guidance and applied to the game-playing systems. Middle [52, 53]. In the field of symbolic regression, MCTS-based methods have been explored as an alternative to genetic programming [54, 55], and researchers have proposed actor–critic style enhancements. Methods to improve search efficiency [56]. Huang et al.'s recent research [57] has further improved MCTS for symbolic regression through enhanced exploration strategies. The most relevant to our work is NEMoTS (Neural Enhancement Monte Carlo Tree Search) [11], which pairs MCTS with policy–value networks for time series symbolic regression. Similarly, AlphaCFG [10] regards alpha discovery as a grammar-guided generation problem, and uses the tree-structured linguistic MDP to force syntactic validity. RiskMiner [58] formalizes alpha mining as a risk-seeking MCTS problem on tokenized expressions at the same time, proving that MCTS-based search can be extended to large-scale financial feature spaces. EATA shares the paradigm of neural-guided search, but the difference is that it clearly reconstructs the guidance signal towards the profit-aware objectives (through the Wasserstein distance), rather than just optimizing IC, and introducing a temporal adaptation mechanism.

More broadly, MCTS has been used as a general mechanism for symbolic and scientific discovery outside the time series, such as the discovery of governing equations [59]. This series of works highlights an important methodological point: search algorithms benefit from domain priors and stable evaluation signals; otherwise, they may waste most of their budgets on exploring areas where syntactically valid but semantically uninformative regions.

The existing symbolic regression method based on MCTS usually optimizes the prediction fit or the accuracy of the benchmark task; the expressions obtained may be explainable, but they are economically weak when deployed as trading rules. Specifically, optimizing MSE can produce accurate predictions near the mean, but the calibration is not good in the tails, which is where trading risks are concentrated.

In addition, two additional boundary conditions are important in the financial market. **(i) Regime dependence:** Expressions found in a time window may decay when volatility, liquidity or macro conditions change. **(ii) Objective mismatch:** Even if a symbolic model predicts the rate of return, a decision layer is needed to convert it into a trading rule; if it is found that the target ignores this decision-making layer, the downstream strategy may be unstable.

EATA solves these problems by coupling neural-guided MCTS with the profit-aware head and using the sliding-window inheritance mechanism to guide the search to those reusable substructures that can survive across regimes.

2.6. Distributional Metrics, Optimal Transport, and Risk-Sensitive Learning

The financial rate of return has the characteristics of heavy-tailed and asymmetric [7], which makes point-wise regression losses (such as MSE) a weak proxy variable of economic utility. Optimal transport provides a geometrically meaningful way to compare distribution, and has been systematically explained in basic literature and modern computational processing methods [60, 61]. Recent work has applied Wasserstein distributionally robust optimization (DRO) to portfolio selection, proving that when the yield distribution is located in the Wasserstein ball around the empirical measure, out-of-sample stability can be achieved [62, 63]. In reinforcement learning, the distributional perspectives [64] and risk-sensitive objectives (such as CVaR optimization [65]) further promote the design that can reflect the tail risk and distributional shape, not just the mean outcomes reward signal. Kastner et al. [66] formalized the distributional model equivalence in risk-sensitive reinforcement learning and established a theoretical connection with modern portfolio theory. The recent work on exponential criteria risk-sensitive RL [67] and risk preference adaptive distributional RL [68] for stock trading further proves the purpose of considering tail risk in financial decision-making. The importance of the target. Zhang [69] proposed a tail-safe hedging strategies that clearly considers extreme events. These insights directly prompt us to use Wasserstein distance as a profit-aware training signal.

Although risk-sensitive criteria (such as CVaR) directly target tail losses, they usually require careful adjustment of tail parameters and may be unstable when tail events are rare. Wasserstein distance provides a complementary mechanism: it compares the entire distribution and is still meaningful in the case of different supports.

The key design problem is not "which indicator is more elegant in theory", but "which indicator can provide a stable learning signal in a heavy-tailed samples". In this sense, Wasserstein distance achieves an inductive bias at the distribution level: it punishes the systematic calibration deviation of quantiles, thus suppressing strategies that perform well in average outcomes but fail in tail events.

2.7. Explainability, Robustness, and Regulation in Finance

In high-risk environments such as the financial field, interpretability is closely related to governance and compliance requirements, including regulatory constraints (such as MiFID II [3]) and broader discussions on explanation rights [70]. Although post-hoc explanation methods such as LIME [35] and SHAP [36] are widely used, their robustness and faithfulness may be questioned, especially in the adversarial manipulations [37]. In addition, there is a continuous debate about whether the attention mechanism constitutes an explanation [71, 72, 73]. These findings support the following view: intrinsically interpretable symbolic strategies may be better in terms of auditability and scientific scrutiny [4, 5, 6].

From the perspective of debate, the core tension exists between *post-hoc transparency* and *intrinsic interpretability*. Post-explanation can provide local insights, but may not meet the robustness or auditability requirements; in contrast, symbolic strategies directly expose decision rule, but there is a risk of overfitting unless it is hypothesis space and objective functions are carefully designed. The design of EATA clearly takes interpretability as a constraint on the hypothesis class (grammar), rather than the visualization layer applied after training.

An emerging supplementary direction is to design models that are self-explaining by construction (not interpreted after training), such as self-explaining neural networks [74]. However, such methods still need to carefully define what is a faithful interpretation, and may not be able to produce a globally auditable rule set. In contrast, symbolic expressions provide explicit, globally applicable decision-making rules that can be checked and stress tested.

2.8. Positioning of EATA

Table 1: Comparison with Related Approaches

Method	Interp.	Neural	Finance	Profit	Stationarity
LSTM/Transformer	✗	✓	✗	✗	✓
FinRL (PPO)	✗	✓	✓	✓	✓
GP (Symbolic)	✓	✗	✗	✗	✗
Alpha Mining [25, 27]	✓	✓	✓	✓	✗
DSR	✓	✓	✗	✗	✗
NEMoTS	✓	✓	✗	✗	✗
EATA (Ours)	✓	✓	✓	✓	✓

Table 1 compares EATA with representative families across key dimensions. Compared to prior alpha mining and symbolic discovery pipelines, EATA uniquely integrates *neural-guided search* (MCTS + PVPNet), *financial operators* (domain grammar), and *profit-aware learning* (distributional reward) under a unified, interpretable framework.

The above literature shows that the core difficulty lies neither in the lack of interpretable models nor in the lack of strong search capabilities, but in the hypothesis space of large-scale, implicit multiple tests, *discovering the mismatch between the objectives and trading utility.* The components of EATA are directly aimed at these failure modes: the profit signal based on Wasserstein targets the distribution-level utility under heavy tails; grammar constraints limit the degrees of freedom and improve auditability; the neural-guided MCTS improves search efficiency while maintaining explicit exploration; sliding-window inheritance provides a practical response mechanism for non-stationarity.

3. Methodology

This section introduces the technical details of the Explainable Algorithmic Trading Agent (EATA). We first establish a unified notation system and then describe the core algorithm and its design principle.

3.1. Problem Formulation

Given a multivariate financial time series $\mathbf{X} = \{\mathbf{x}_t\}_{t=1}^{T}$ where $\mathbf{x}_t \in \mathbb{R}^d$ represents d market features at time t, we seek an interpretable expression e^* that balances predictive accuracy with trading profitability:

$$e^* = \arg\max_{e \in \mathcal{E}} \left[\eta \cdot \mathcal{A}(e, \mathbf{X}) + (1 - \eta) \cdot \mathcal{P}(e, \mathbf{X})\right], \quad (1)$$

where $\mathcal{E}$ denotes the valid expression space defined by grammar $\mathcal{G}$, $\mathcal{A}(\cdot)$ measures prediction accuracy, $\mathcal{P}(\cdot)$ measures trading profit, and $\eta \in [0, 1]$ balances the two objectives.

Rationale. It is known that the pure accuracy objective is inconsistent with the trading utility, because it handles all errors symmetrically and ignores the asymmetry of risk. On the contrary, the profit objective may overfit return realizations and produce brittle expressions that takes advantage of idiosyncratic fluctuations. The mixed term in Eq. 1 plays the role of a regularization mechanism: $\mathcal{A}$ will stabilize the search on the expression that can capture the repeatable predictable structure, while $\mathcal{P}$ makes the discovered expression biased To the results of economic significance. Therefore, the control parameter η reveals a transparent trade-off between statistical fit and economic utility, which can be adjusted to risk preferences.

3.2. Domain-Specific Grammar Design

We employ a context-free grammar $\mathcal{G} = (\mathcal{V}, \Sigma, \mathcal{R}, S)$ tailored for financial time series analysis. The grammar incorporates domain knowledge through fixed-window financial operators.

Non-terminals and Terminals. The non-terminal set $\mathcal{V} = \{S, A, B, D\}$ includes the start symbol (S), primary expressions (A), helper expressions (B), and decision predicates (D). The terminal set Σ comprises feature variables $\{x_0, x_1, \ldots, x_{d-1}\}$, constants, and operators.

Production Rules. The rule set $\mathcal{R}$ comprises the start rule and four operator categories, including:

1) Start Rule

$$S \to A \quad (2)$$

2) Basic Arithmetic

$$A \to A + A \mid A - A \mid A \times A \mid A \div A$$
$$\mid A \times C \mid A + C \quad (3)$$

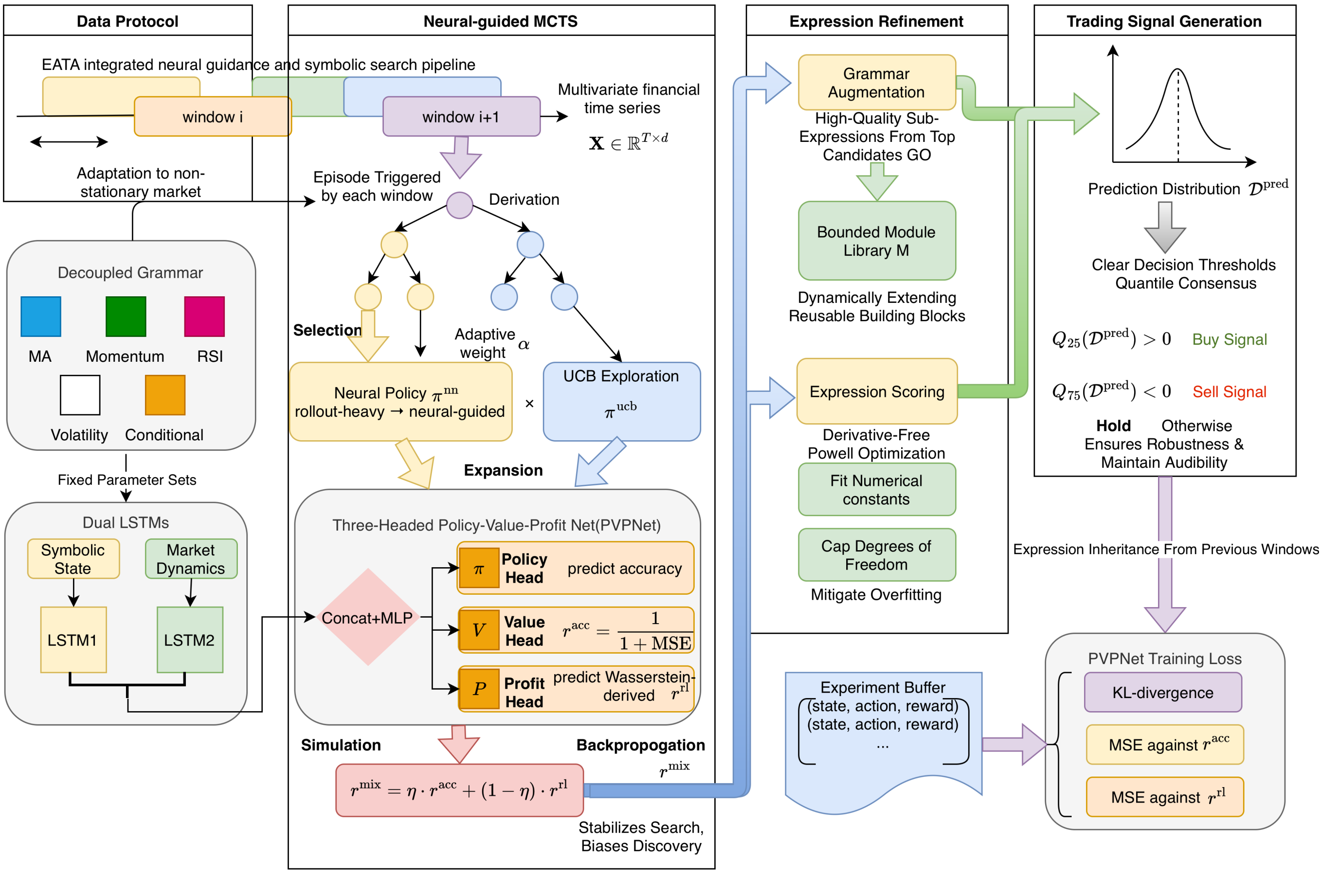

Figure 1: EATA Integrated Neural Guidance and Symbolic Search Pipeline. The framework operates through four main components: (1) **Data Protocol**: Sliding window adaptation with decoupled grammar (MA, Momentum, RSI, Volatility, Conditional operators); (2) **Neural-Guided MCTS**: Episode-triggered derivation with selection, expansion, and adaptive weight α balancing neural policy π_{nn} and UCB exploration π_{ucb}; (3) **Expression Refinement**: Grammar augmentation via bounded module library, expression scoring with derivative-free Powell optimization, and numerical constant fitting; (4) **Trading Signal Generation**: Three-headed PVPNet (Policy-Value-Profit) with dual LSTMs encoding symbolic states and market dynamics, profit-aware reward via 1-Wasserstein distance aligning predicted and actual return distributions, and quantile-based consensus signals (Q25/Q75 thresholds for Buy/Sell decisions).

3) Mathematical Functions

$$A \to \cos(A) \mid \sin(A) \mid \exp(A) \mid \log(B) \mid \sqrt{B} \mid |A| \mid A^2 \tag{4}$$

4) Financial Operators (Decoupled Parameters):

$$A \to \mathrm{delay}(A, L) \mid \mathrm{ma}(A, W) \mid \mathrm{diff}(A, L) \mid \mathrm{mom}(A, L) \mid \mathrm{max\_n}(A, W) \mid \mathrm{min\_n}(A, W) \mid \mathrm{rsi}(A, W) \mid \mathrm{vol}(A, W) \tag{5}$$

$$W \to 5 \mid 10 \mid 20 \mid 30 \mid 60, \quad L \to 1 \mid 5 \mid 10 \mid 20 \tag{6}$$

5) Conditional Logic

$$A \to \mathrm{ite}(D, A, A), \quad D \to A > B \mid A < B \tag{7}$$

6) Helper Expressions (Guaranteed Non-Negative Output):

$$B \to |A| \mid A^2 \mid \mathrm{ma}(A, W) \mid \mathrm{vol}(A, W) \mid \mathrm{max\_n}(A, W) \mid \mathrm{min\_n}(A, W) \tag{8}$$

Design Rationale. We adopt a decoupled grammar design inspired by AlphaCFG [10], separating operators from their parameters. This introduces parameter non-terminals (W for windows, L for lags) that resolve to a discrete set of valid values. This compositional structure reduces the number of unique production rules while maintaining expressiveness, and prevents the generation of semantically invalid parameter combinations (e.g., negative windows). The conditional operator ite enables piecewise expressions for regime-switching strategies.

Rationale and Alternatives. In finance, unconstrained operator sets often cause expressions to be either uninterpretable(deeply nested non-linearities) or non-robust (operators "search" for hyperparameters through continuous constants). Fixed-window operators have two functions: they encode domain priors and reduce the multiple-testing effects by limiting the degrees of freedom. In contrast, allowing arbitrary window lengths will significantly expand the hypothesis space and increase the risk of data snooping [16, 18]. Grammar constraints can be regarded as a structural regularizer, which can improve both interpretability and generalization.

3.3. Neural-Guided Monte Carlo Tree Search

Algorithm 1 presents the MCTS procedure with neural network guidance. Each episode traverses the search tree using a policy fusion mechanism.

Design Rationale. The dual fusion mechanism solves the uncertainty of neural network predictions in the early stage of training. When α is smaller (the buffer is not full), the rollout provides reliable value estimates; as the training progresses, α increases, and the trust in the neural network also increases. Value-profit fusion through ω ensures that expressions take into account both predictive fit and profit potential when selecting.

Why MCTS (vs. GP / sequence generators)? GP is effective, but it may encounter problems of bloat and high cost of evaluation, especially when using backtesting-based objectives. The serial generator (RNN/Transformer) has good scalability, but usually requires a large amount of training data, and may generate expressions that are syntactically valid but semantically meaningless expressions without strong constraints. MCTS provides a controllable exploration mechanism: it can use neural priors when information is available, while retaining explicit exploration through UCB. This is especially applicable to financial time series, because its signal-to-noise ratio is relatively low, and naive exploitation may lead to overfitting.

3.4. Expression Scoring with Coefficient Optimization

Algorithm 2 evaluates expressions and optimizes numerical constants.

Rationale and Alternatives. Constant fitting is indispensable, because grammar encodes the functional forms, and the numerical coefficients determine the scale and sensitivity. If there is no coefficient optimization, the search will either discard the correctly structured expression because of poor initialization constants, or expand the grammar to contain many approximately repeated scaling variants, thus increasing the hypothesis space. We use a derivative-free optimizer (Powell) because expression evaluation may be non-smooth (for example, due to the existence of conditional operators) and the gradient is not easy to obtain. The constant cap alleviates a common failure mode in symbolic regression: degrees of freedom may produce brittle expressions that fit the noisy targets. Other optional methods include linear least squares for the linear-in-parameters forms, or gradient-based optimization for differentiable operators; our choice prioritizes robustness in heterogeneous expression structures.

3.5. Grammar Augmentation via Module Extraction

Algorithm 3 extracts high-quality sub-expressions.

Rationale and Alternatives. Module extraction is a bias–variance trade-off in symbolic discovery. From the perspective of search, reusing high-quality sub-expressions

Algorithm 1 Neural-Guided MCTS for Symbolic Regression

Require: Data $\mathbf{X}$, grammar $\mathcal{G}$, network f_θ, episodes K, guidance weight α, fusion weight ω, prior tree τ^{prev}
Ensure: Best expression e^*, best tree τ^*, experience buffer $\mathcal{D}$

Initialize $Q(\cdot) \leftarrow 0$, $N(\cdot) \leftarrow 1$, $\mathcal{D} \leftarrow \emptyset$
$s \leftarrow \textsc{InitializeState}(\tau^{\text{prev}})$
for $k = 1$ **to** K **do**
  states $\leftarrow$ [], *actions* $\leftarrow$ []
  $\mathcal{U} \leftarrow \textsc{GetUnvisited}(s)$
  while $\mathcal{U} = \emptyset$ **and** not $\textsc{IsTerminal}(s)$ **do** ▷ Selection
    $\boldsymbol{\pi}^{\text{nn}}, V, P \leftarrow f_\theta(s, \mathbf{X})$
    $\boldsymbol{\pi}^{\text{ucb}} \leftarrow \textsc{ComputeUCB}(s, C)$
    $\boldsymbol{\pi} \leftarrow \alpha \cdot \boldsymbol{\pi}^{\text{nn}} + (1 - \alpha) \cdot \boldsymbol{\pi}^{\text{ucb}}$
    $\boldsymbol{\pi} \leftarrow \textsc{Normalize}(\boldsymbol{\pi})$
    $a \sim \text{Categorical}(\boldsymbol{\pi})$
    states.append(s), *actions*.append(a)
    $s \leftarrow \textsc{Step}(s, a)$
    $\mathcal{U} \leftarrow \textsc{GetUnvisited}(s)$
    if $|s| \geq L^{\max}$ **then**
      $\textsc{Backprop}(\textit{states}, \textit{actions}, 0)$
      break
    end if
  end while
  if $\mathcal{U} \neq \emptyset$ **then** ▷ Expansion
    $\boldsymbol{\pi}^{\text{nn}}, V, P \leftarrow f_\theta(s, \mathbf{X})$
    $a \sim \text{Categorical}(\alpha \cdot \boldsymbol{\pi}^{\text{nn}} + (1-\alpha) \cdot \textsc{Uniform}(\mathcal{U}))$
    $s' \leftarrow \textsc{Step}(s, a)$
    $r^{\text{acc}}, e \leftarrow \textsc{Evaluate}(s', \mathbf{X})$ ▷ Accuracy reward via Alg. 2
    if $\textsc{IsTerminal}(s')$ **then**
      $\textsc{UpdateModules}(e, r^{\text{acc}})$
    end if
  else ▷ Simulation via leaf evaluation
    $r^{\text{acc}}, e \leftarrow \textsc{Evaluate}(s, \mathbf{X})$ ▷ Accuracy reward via Alg. 2
  end if
  $r^{\text{rl}} \leftarrow \textsc{ComputeRLReward}(e, \mathbf{X})$ ▷ Profit reward via Eq. 10
  $v^{\text{nn}} \leftarrow \omega \cdot V + (1 - \omega) \cdot P$ ▷ Fuse value heads
  $v^{\text{roll}} \leftarrow \textsc{Rollout}(s, n^{\text{play}})$ ▷ Random simulations
  $r^{\text{final}} \leftarrow \alpha \cdot v^{\text{nn}} + (1 - \alpha) \cdot v^{\text{roll}}$ ▷ Fuse neural and rollout
  $\textsc{Backprop}(\textit{states}, \textit{actions}, r^{\text{final}})$
  $\mathcal{D}$.append(($\textit{states}, \mathbf{X}, \boldsymbol{\pi}, r^{\text{acc}}, r^{\text{rl}}$))
end for
return $e^*, \tau^*, \mathcal{D}$

Algorithm 2 Expression Scoring with Coefficient Estimation

Require: Expression e, data $(\mathbf{X}, \mathbf{y})$
Ensure: Score r, optimized expression e^{opt}

$n_c \leftarrow \textsc{CountConstants}(e)$
if $n_c = 0$ **then**
    $\hat{\mathbf{y}} \leftarrow \textsc{Evaluate}(e, \mathbf{X})$
else if $n_c \geq 10$ **then**
    return $0, e$ ▷ Too many constants
else
    $\mathbf{c}^* \leftarrow \arg\min_{\mathbf{c}} \|\textsc{Evaluate}(e, \mathbf{X}, \mathbf{c}) - \mathbf{y}\|_2$ ▷ Powell optimization, timeout $t^{\max} = 30\text{s}$
    $e^{\text{opt}} \leftarrow \textsc{Substitute}(e, \mathbf{c}^*)$
    $\hat{\mathbf{y}} \leftarrow \textsc{Evaluate}(e^{\text{opt}}, \mathbf{X})$
end if
$\text{MSE} \leftarrow \frac{1}{T}\|\hat{\mathbf{y}} - \mathbf{y}\|_2^2$
$r \leftarrow \frac{1}{1+\text{MSE}}$ ▷ Accuracy reward
return r, e^{opt}

Algorithm 3 Grammar Augmentation

Require: Module library $\mathcal{M}$, expression e, reward r, max length L^{mod}, max size $M^{\max}$
Ensure: Updated library $\mathcal{M}'$

$\mathcal{M}' \leftarrow \mathcal{M}$
for each sub-expression $m \in \textsc{Extract}(e)$ **do**
    if $\textsc{Length}(m) \leq L^{\text{mod}}$ **and** $m \notin \mathcal{M}'$ **then**
        if $|\mathcal{M}'| < M^{\max}$ **then**
            $\mathcal{M}' \leftarrow \mathcal{M}' \cup \{(m, r)\}$
        else if $r > \min_{(m', r') \in \mathcal{M}'} r'$ **then**
            $\mathcal{M}' \leftarrow \mathcal{M}' \setminus \{(m^{\min}, r^{\min})\}$
            $\mathcal{M}' \leftarrow \mathcal{M}' \cup \{(m, r)\}$
        end if
    end if
end for
return $\mathcal{M}'$

can shorten the effective depth required to construct excellent candidate expressions, thus improving sample efficiency. From the perspective of generalization, a bounded module library can prevent uncontrolled bloat and limit the number of reusable building blocks, thus reducing the risk of idiosyncratic patterns. Other options include: maintaining a unbounded library (the risk of overfitting and redundancy is higher), using the library learning style of probabilistic program induction, or learning the continuous embeddings of modules; we use explicit, bounded module libraries to maintain interpretability and maintain transparent search space.

3.6. Profit-Aware RL Reward via Wasserstein Distance

In order to explicitly optimize the trading profitability, we have introduced Reinforcement Learning (RL) reward signal r^{rl}. Unlike the standard regression target of punishing point-wise errors (such as MSE), financial transactions need to capture the distributional properties of returns, especially the asymmetry between gains and losses and the existence of heavy tails. This is consistent with the broader trend in distributional and risk-sensitive reinforcement learning. These methods model the return distributions or tail risk objectives, not just expected value [64, 65].

Our key insight is that **Wasserstein distance** (Earth Mover's Distance) provides a theoretically based way to measure the yield distribution, which takes into account tail risk and directional asymmetry at the same time [60, 61]. Unlike KL divergence or MSE, Wasserstein distance measures the minimum "work" required to transform one distribution into another, which makes it sensitive to the magnitude and location of distributional differences.

Given predictions $\hat{\mathbf{y}}^{(1)}, \ldots, \hat{\mathbf{y}}^{(k)}$ from the top-k discovered expressions, we construct a prediction distribution $\mathcal{D}^{\text{pred}}$. We compare this against the distribution of actual future returns $\mathcal{D}^{\text{actual}}$ using the 1-Wasserstein distance (Earth Mover's Distance) [60, 61]:

$$\mathcal{W}_1(\mathcal{D}^{\text{pred}}, \mathcal{D}^{\text{actual}}) = \int_{-\infty}^{\infty} |F^{\text{pred}}(x) - F^{\text{actual}}(x)| \, dx \quad (9)$$

where F^{pred} and F^{actual} are the cumulative distribution functions (CDFs) of the predicted and actual returns, respectively. The RL reward is then defined as:

$$r^{\text{rl}} = \frac{1}{1 + \mathcal{W}_1(\mathcal{D}^{\text{pred}}, \mathcal{D}^{\text{actual}})} \quad (10)$$

Why Wasserstein? The return distributions are essentially non-Gaussian, usually showing skewness and leptokurtosis. For this kind of data, Wasserstein distance has a significant theoretical advantage over Kullback-Leibler (KL) divergence or MSE: (1) Even if the support sets of the two distributions do not overlap, it can provide a meaningful measure; (2) It considers space The geometry of the space (that is, the size of the error is very important, not just the difference in probability quality); (3) Mapping the predicted distribution to the transport cost of the actual distribution by measuring, it can effectively capture the tail risks [60, 61]. This ensures that the intelligent body will be punished not only for the error in the mean prediction, but also for failing to model the risk profile of the asset. In addition to these theoretical advantages, recent studies also show that Wasserstein DRO can improve the out-of-sample stability in portfolio selection [62, 63], which provides an empirical motivation for it as a trading-oriented objective. Appendix Appendix C provides a formal analysis of the gradient properties, convergence guarantees and CVaR-control relationship of Wasserstein reward.

3.6.1. Reward Semantics, Training Targets, and Objective Variants

EATA uses two conceptually distinct signals during search and training. **(i) Accuracy reward** evaluates point-wise prediction fit, consistent with conventional regression objectives:

$$r^{\mathrm{acc}} = \frac{1}{1 + \mathrm{MSE}}. \tag{11}$$

(ii) Profit-aware reward evaluates distributional alignment between predicted and realized return distributions via Wasserstein distance, as defined in Eq. 10.

Data flow. During MCTS, each terminal expression is evaluated to obtain $(r^{\mathrm{acc}}, r^{\mathrm{rl}})$; the search backup uses a scalar reward that prioritizes profit while retaining an optional accuracy component:

$$r^{\mathrm{mix}} = \eta \cdot r^{\mathrm{acc}} + (1 - \eta) \cdot r^{\mathrm{rl}}. \tag{12}$$

This formulation avoids degenerate expressions that exploit noisy returns: r^{acc} stabilizes the search toward repeatable predictive structure, while r^{rl} biases discovery toward economically meaningful tail behavior.

PVPNet targets. The value head V_θ is trained to predict r^{acc}, while the profit head P_θ is trained to predict r^{rl}. This separation makes the guidance signal interpretable: the improvement of P_θ corresponds to the discovery of expressions with better distributional risk-return characteristics, not just lower MSE.

RQ1 baseline (MSE-based objective). In order to isolate the effect of the profit-aware objective, we define accuracy-only variant, written as **NoRL** (or **EATA-MSE**), which removes the profit head and replaces the profit perception reward with the accuracy reward, that is, $r^{\mathrm{rl}} \leftarrow r^{\mathrm{acc}}$ and $r^{\mathrm{mix}} \leftarrow r^{\mathrm{acc}}$. The variant retains the same grammar, search budget and temporal protocol, so that it can provide a controlled comparison for RQ1.

3.7. Three-Headed Policy-Value-Profit Network (PVPNet)

The PVPNet architecture (Algorithm 4) processes both the derivation state and input time series.

The network is trained via three-component loss:

$$\mathcal{L}(\theta) = \lambda^{\pi}\mathcal{L}^{\pi} + \lambda^{V}\mathcal{L}^{V} + \lambda^{P}\mathcal{L}^{P} \tag{13}$$

Algorithm 4 PVPNet Forward Pass

Require: State sequence $\boldsymbol{\sigma} = [r_1, \ldots, r_l]$, time series $\mathbf{X} \in \mathbb{R}^{T \times d}$

Ensure: Policy $\boldsymbol{\pi}$, value V, profit P

1: $\mathbf{E} \leftarrow \textsc{Embedding}(\boldsymbol{\sigma})$ ▷ Rule embeddings
2: $\mathbf{h}^{\mathrm{state}} \leftarrow \mathrm{LSTM}_{\mathrm{state}}(\mathbf{E})$ ▷ State encoder
3: $\mathbf{h}^{\mathrm{seq}} \leftarrow \mathrm{LSTM}_{\mathrm{seq}}(\mathbf{X})$ ▷ Sequence encoder
4: $\mathbf{h} \leftarrow [\mathbf{h}^{\mathrm{state}}; \mathbf{h}^{\mathrm{seq}}]$ ▷ Concatenate
5: $\mathbf{h} \leftarrow \mathrm{MLP}(\mathbf{h})$ ▷ Hidden representation
6: $\boldsymbol{\pi} \leftarrow \textsc{Softmax}(\mathbf{W}^{\pi}\mathbf{h} + \mathbf{b}^{\pi})$
7: $V \leftarrow \mathbf{w}^{V\top}\mathbf{h} + b^{V}$
8: $P \leftarrow \mathbf{w}^{P\top}\mathbf{h} + b^{P}$
9: **return** $\boldsymbol{\pi}, V, P$

where the components are:

$$\mathcal{L}^{\pi} = D_{\mathrm{KL}}(\boldsymbol{\pi}^{\mathrm{target}} \| \boldsymbol{\pi}_\theta) \tag{14}$$
$$\mathcal{L}^{V} = (V_\theta - r^{\mathrm{acc}})^2 \tag{15}$$
$$\mathcal{L}^{P} = \left(P_\theta - r^{\mathrm{rl}}\right)^2 \tag{16}$$

Design Rationale. Dual-LSTM design encodes the symbolic derivation path (state LSTM) and market dynamics (sequence LSTM) respectively. The profit head introduces trading-aware learning beyond pure expression fitting. Both the value head and the profit head use MSE loss to fit their respective reward signals, of which the accuracy reward is $r^{\mathrm{acc}} = 1/(1+\mathrm{MSE})$, and the profit reward r^{rl} is based on Wasserstein distance is derived.

On the compact hidden dimension. The hidden dimension used by PVPNet is $d_h = 16$, which is deliberately set smaller. The reason why it is feasible is that MCTS significantly reduces the effective search space: the grammar constraints and module library limit the branching factor to about $|\mathcal{R}| \approx 30$ of production rules, and the policy head only needs to select these discrete rules in each derivation step. Item to distinguish. Experiments show that the benefits of increasing d_h to 32 are negligible (see Appendix **??**), which confirms that 16-dimensional is sufficient for this constrained search setting.

3.8. Overall Training Pipeline

Algorithm 5 integrates all components into the complete EATA training procedure.

Design Rationale. Sliding windows can adapt to non-stationary markets. Expression inheritance (using τ^{prev} warm-starting MCTS) to speed up search by reusing high-quality partial derivation. With the accumulation of experience, adaptive α schedule transitions from the exploration (rollout-heavy) to the exploitation (NN-guided).

Rationale and Alternatives. There are structural breaks and regime shifts in the financial market; training a single static expression on a long window usually leads to a decline in the performance of the strategy when the volatility and correlations change. The sliding window provides a practical compromise between full retraining

Algorithm 5 EATA Training Pipeline

Require: Training data $\mathbf{X}$, hyperparameters $\{T^{\text{in}}, H, K, M, N^{\text{run}}\}$
Ensure: Trained network f_θ, best expression e^*

Initialize f_θ, $\mathcal{B} \leftarrow \emptyset$, $\mathcal{G}$, $\tau^{\text{prev}} \leftarrow$ NULL
for each sliding window w **do**
  for $i = 1$ **to** N^{run} **do**
    $\mathcal{M} \leftarrow \emptyset$
    for $j = 1$ **to** M **do**
      $\mathcal{G}' \leftarrow \mathcal{G} \cup \{A \to m : (m, \cdot) \in \mathcal{M}\}$
      $\alpha \leftarrow \min(1, |\mathcal{B}|/B^{\max})$ ▷ Adaptive guidance
      $e, \tau, \mathcal{D} \leftarrow \text{MCTS}(\mathbf{X}_w, \mathcal{G}', f_\theta, K, \alpha, \omega, \tau^{\text{prev}})$
      $\mathcal{M} \leftarrow \textsc{Augment}(\mathcal{M}, e)$
    end for
  end for
  $r^{\text{rl}} \leftarrow \textsc{ComputeRLReward}(e, \mathbf{X}_w)$ ▷ Via Eq. 10
  for $(s, \mathbf{x}, \boldsymbol{\pi}, v) \in \mathcal{D}$ **do**
    $\mathcal{B} \leftarrow \mathcal{B} \cup \{(s, \mathbf{x}, \boldsymbol{\pi}, v, r^{\text{rl}})\}$
  end for
  if $|\mathcal{B}| > B^{\min}$ **then**
    Sample mini-batch from $\mathcal{B}$
    Update θ by minimizing $\mathcal{L}$ (Eq. 13)
  end if
  $\tau^{\text{prev}} \leftarrow \tau$ ▷ Expression inheritance
end for
return f_θ, e^*

(high computing cost and instability) and freezing model (non-adaptive). The motivation of inheritance comes from an empirical observation: even when the coefficient or threshold value changes, useful substructures (such as the moving-average trend filters) still persist between adjacent windows. Therefore, Warm-starting the tree can reduce redundant searches and improve stability. Alternatives include online learning with continuous parameter updates, meta-learning for quick adaptation, or the use of multiple expert models for regime detection; EATA adopts inheritance and windowing methods because they can maintain interpretability (explicit expressions) while providing a simple and auditable adaptation mechanism.

3.9. Trading Signal Generation

Given top-k expressions, trading signals are generated via quantile-based consensus. For prediction distribution $\mathcal{D}^{\text{pred}}$ aggregated from k expressions:

$$\text{signal} = \begin{cases} +1 \text{ (buy)} & \text{if } Q_{25}(\mathcal{D}^{\text{pred}}) > 0 \\ -1 \text{ (sell)} & \text{if } Q_{75}(\mathcal{D}^{\text{pred}}) < 0 \\ 0 \text{ (hold)} & \text{otherwise} \end{cases} \tag{17}$$

Design Rationale. The $Q_{25} > 0$ rule ensures a strong bullish consensus before establishing long positions (even the pessimistic 25th percentile predicts positive returns). On the contrary, $Q_{75} < 0$ ensures a strong bearish consensus before shorting. Compared with median-based rules, this conservative threshold reduces false signals. These specific contals are selected based on the performance of the validation set and aim to prioritize signal precision over the recall, because the cost of false positives in trading is high.

Rationale and Alternative. A common failure mode in predictive trading is that weak signals near zero will produce churn, thus amplifying trading costs. Quantile-based consensus provides a simple and explainable robustness mechanism: it requires distribution-level agreement between candidate expressions and acts as a "margin" against noisy predictions. Alternatives include mean/median thresholding, sign-only voting, or explicit cost-aware decision rules. We adopt quantile consensus because it directly reflects the uncertainty of the discovered expression set while maintaining interpretability (a clear, auditable decision boundary).

3.10. Complexity Analysis

Time Complexity. Each MCTS episode traverses $O(L^{\max})$ nodes. With K episodes, M augmentation iterations, and N^{run} independent runs per window, the complexity is $O(N^{\text{run}} \cdot M \cdot K \cdot L^{\max})$. Network inference adds $O(|\mathcal{R}|)$ per decision.

Space Complexity. The MCTS tree stores $O(K \cdot L^{\max})$ nodes. The experience buffer is bounded by $B^{\max}$. The module library is bounded by $M^{\max}$.

4. Experiments

This section conducts a comprehensive evaluation of EATA and answers the research questions raised in the 1 section. We evaluate on the **representative constituent stocks** data set (N=62 stocks) in the S&P 500 index, covering multiple industries such as Technology, Finance and Healthcare. The evaluation time span is from **February 8, 2013 to February 7, 2018**, covering a variety of market regimes:

- **2013-2014**: Post-financial crisis recovery and QE tapering.
- **2015**: Oil price collapse and emerging market concerns.
- **2016**: Brexit and U.S. presidential election volatility.
- **2017**: Strong bull market with low volatility.
- **2018**: Fed tightening and early-year correction.

Daily OHLCV data is split temporally: training period (Feb 8, 2013 to Jun 30, 2016) and out-of-sample testing period (Jul 1, 2016 to Feb 7, 2018).

4.1. Baselines

We compare EATA against 11 baselines across four categories:

- **Traditional**: Buy & Hold, MACD (tuned) [75].
- **Statistical**: ARIMA (5,1,0).
- **Machine Learning**: LightGBM [33], GBDT, Genetic Programming (GP, via Operon) [41].
- **Deep Learning**: LSTM [1, 76], Transformer [28, 29] .
- **Reinforcement Learning**: PPO, A2C, TD3, DDPG [77] (FinRL implementation [31]).

Detailed hyperparameter settings are provided in Appendix Appendix A.

4.2. Main Results: Profitability vs. Black-Box (RQ3)

4.2.1. Performance Comparison

Hypothesis (RQ3). Profit-aware symbolic strategies can match or exceed black-box baselines in risk-adjusted returns while remaining intrinsically interpretable.

Table 2 reports performance metrics averaged across our evaluation universe. EATA achieves competitive risk-adjusted returns with a **Sharpe Ratio of 0.76**, demonstrating strong performance while maintaining full interpretability through symbolic expressions.

Table 2: Main Performance Comparison. EATA achieves competitive Sharpe Ratio while maintaining interpretability.

Method	AR (%)	SR	MDD (%)	Win Rate	Calmar
Buy & Hold	12.7	0.57	12.0	55%	1.05
PPO (FinRL)	7.5	0.56	8.4	58%	0.89
A2C (FinRL)	7.1	0.47	8.6	52%	0.83
MACD	6.0	0.40	9.3	50%	0.65
DDPG (FinRL)	5.4	0.34	6.4	31%	0.85
TD3 (FinRL)	3.9	0.27	7.1	32%	0.54
Transformer	0.0	-0.03	4.2	24%	0.01
ARIMA	-2.5	-0.12	21.3	32%	-0.12
GBDT	-4.5	-0.22	19.7	21%	-0.23
GP (Operon)	-4.7	-0.22	19.3	23%	-0.24
LSTM	-11.0	-0.59	25.5	10%	-0.43
EATA (Ours)	**21.2****	**0.76****	36.1	51.8%	0.62

4.3. Threats to Validity and Mitigations

Multiple testing and data snooping. Whenever a large number of candidate expressions, operators or hyperparameter settings are explored, the probability of accidentally finding seemingly profitable strategies increases. Classic reality-check style procedures [16, 17] and multiple test analysis in asset pricing [20] pointed out that naive p-values may be too optimistic. Therefore, we regard the importance tests of the report as indicative, and emphasize controlled ablations and robustness checks; for the conclusion of the whole stock pool, a more conservative multiple test correction should be adopted.

Backtest overfitting. Modern analysis shows that even if each individual backtest is an appropriate out-of-sample test, the backtest selection itself may introduce fitting, especially when many variants are tried and the best results are reported [18, 19]. This is especially relevant for symbolic search, because the discovery process itself is an implicit multiple-hypothesis test. Our design structurally alleviates this risk through grammar constraints and the separation of profit-aware objective and point-wise fitting.

Metric uncertainty under heavy tails. Risk-adjusted indicators such as the Sharpe ratio may be noisy and sensitive to non-Gaussianity [14, 15]. In order to make rigorous inferences, uncertainty estimates (such as the block bootstrap) should be reported and the tail-aware risk measures should be considered. Therefore, our mechanism analysis not only focuses on average performance, but also on tail-related quantities (MDD/Calmar) and distributional diagnostics.

We follow the classical portfolio theory [12, 13] and report the Sharpe ratio as the main risk-adjusted indicator.

The consistency advantage of EATA over symbolic baselines (genetic programming) and black-box predictors (LightGBM, LSTM, Transformer) shows that it is crucial to optimize the discovery process towards *trading utility*. We attribute the benefits to two coupled design options: (i) the profit perception target that punishes the distributional risk characteristics, and (ii) the domain grammar that constrains the search on the operator of financial significance. In contrast, purely predictive depth models are prone to overfitting in low signal-to-noise regimes and may produce statistically accurate but economically weak predictions.

4.3.1. Cumulative Returns

Figure 2 compares the cumulative returns of EATA against key baselines. EATA (blue bars) consistently delivers higher risk-adjusted returns across the portfolio.

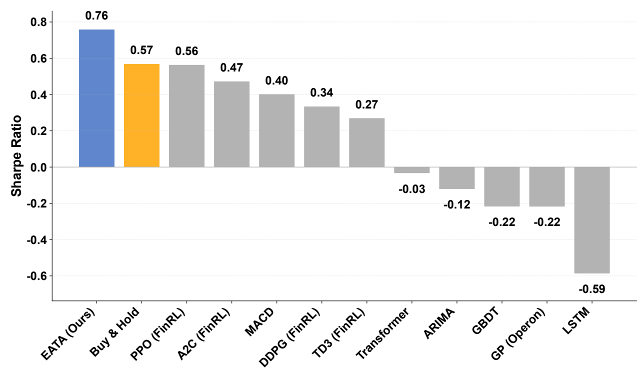

Figure 2: Cumulative Performance Comparison. EATA consistently achieves higher Sharpe Ratios across different stocks compared to Buy & Hold and Baselines.

Discussion (RQ3). We emphasize that the cumulative-return curves should be interpreted in combination with the risk statistics. In particular, the two strategies may achieve similar final returns, but there are significant differences in path-wise risk (drawdowns) and volatility. In our comparison, EATA shows a smoother growth trajectory than several baselines, which shows that the profit-aware objective not only increases the mean return, but also implicitly shapes the return path. This observation is consistent with the premise of this article: the accuracy of point-wise prediction accuracy cannot control the economically relevant tail outcomes, while the distribution-aware objective provides a more direct training signal for the quality of risk-return.

4.3.2. Strategy Correlation

Figure 3 shows the correlation between EATA's earnings and other strategies. EATA has low correlation with traditional trend-following strategies, which shows that it captures unique alpha sources that standard technical indicators have not yet been used.

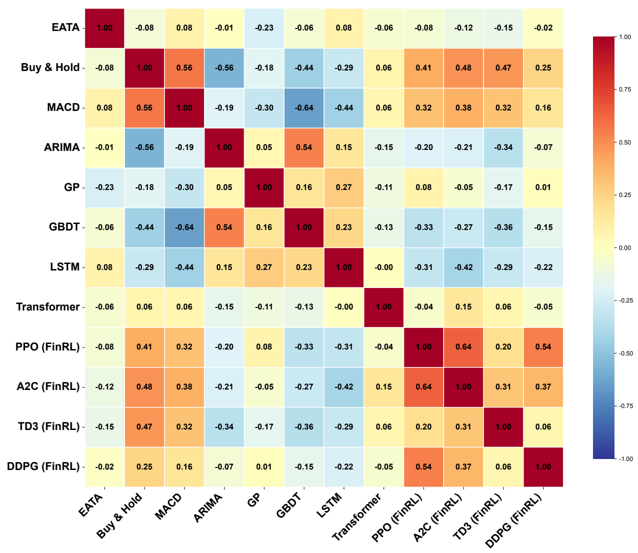

Figure 3: Strategy Correlation Matrix. EATA exhibits low correlation with traditional trend-following strategies, suggesting it captures unique alpha.

Discussion (RQ3). Correlation analysis has two purposes. First of all, the low correlation with Buy & Hold shows that the strategy is not simply repackaging the market beta. Secondly, the low correlation with other learning baselines indicates that EATA may extract a set of unique predictive patterns (for example, a momentum/mean-reversion combination that regime-dependent). From the perspective of portfolio , even if the per-asset Sharpe ratios are similar, this low correlation can be transformed into diversification benefits. However, correlation estimates are noisy in finite samples and may be unstable under regime shifts; therefore, the analysis should be regarded as supportive evidence rather than an independent conclusion.

4.4. Mechanism Analysis: Wasserstein vs. MSE (RQ1)

4.4.1. Return Distribution

Hypothesis (RQ1). Optimizing distributional divergence (Wasserstein) instead of point-wise error (MSE) can improve tail-risk control and produce a return distribution with more favorable asymmetry.

Figure 4 shows the return distribution generated by EATA. The distribution shows positive skew (skewness = 1.14), indicating that EATA can achieve occasional high-return out-of-group values while controlling downside risks. This asymmetry is consistent with the Wasserstein-based profit objective, which punishes the distributional mismatches. The mean daily return of 0.076% corresponds to the annualized return of 21.2% reported in Table 2. Quantitatively, Table 3 provides a controlled comparison with the MSE-based NoRL variant defined in Section 3.6.1.

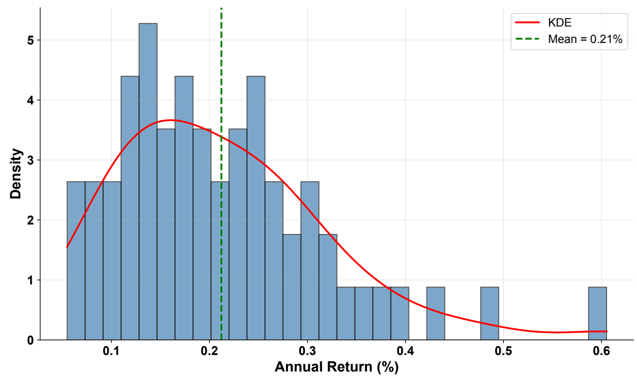

Figure 4: Return Distribution. EATA's return distribution shows a positive skew (skewness = 1.14, mean = 0.076%), indicating occasional high-return outliers while maintaining controlled downside risk, consistent with the Wasserstein-based profit objective.

Discussion (RQ1). The key difference between distribution perception objectives and point-wise objectives is that the former punishes mismatches in the entire return distribution (including extreme quantiles), while MSE is mainly dominated by frequent small errors near the center. In trading, due to the drawdown constraints and leverage effects, the left tail (loss events) may dominate the realized utility. Therefore, even if the average point prediction error is not minimized, the distribution perception target is expected to produce fewer catastrophic losses. This provides a mechanical explanation for the ability of the Profit Head to reduce the tail risk in the Table 3.

4.4.2. Risk-Return Trade-off

Figure 5 visualizes the Risk-Return profile of all stocks. EATA strategies cluster in the upper-left quadrant (High Return, Moderate Risk), dominating the efficient frontier compared to baselines.

Discussion (RQ1/RQ3). The risk-return scatter plots helps to clarify whether performance gains is achieved by taking additional risks or by improving risk-adjusted efficiency. The favorable shift to higher returns at similar

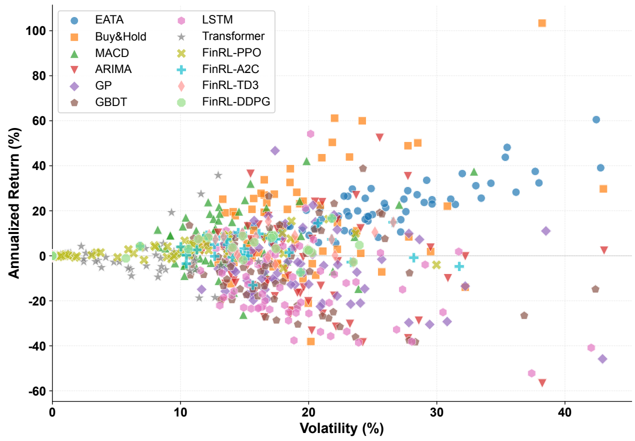

Figure 5: Risk-Return Scatter Plot. Each point represents a stock. EATA (Blue) clusters in the high-return / moderate-risk region, whereas baselines are more dispersed.

volatility is consistent with the idea that the target function will align the search with economically meaningful outcomes. We remind that under heavy tails, the volatility itself alone is not a complete risk metric [7]; however, combined with the MDD/Calmar in the Table 2 Indicators, this perspective helps to distinguish between "high-yield but unstable" strategies and truly risk-efficient strategies.

4.4.3. Ablation Study: Component Analysis

Table 3: Ablation Study Results: Performance Metrics by Variant

Configuration	AR (%)	SR	MDD (%)	Win Rate
EATA (Full)	**21.2**	**0.76**	**36.1**	**51.8%**
w/o MCTS	4.3	0.19	30.9	50.4%
w/o Exploration	9.5	0.43	31.1	51.1%
w/o Memory	3.1	0.15	34.8	44.2%
w/o Neural Network	10.6	0.45	31.4	51.1%
w/o Wasserstein	4.1	0.22	37.3	48.5%

Finding (RQ1: Component Importance). The ablation study reveals a clear hierarchy of component importance. **MCTS is the most critical component**: Removing it will lead to a significant decrease in the Sharpe Ratio (0.76 $\rightarrow$ 0.19, -75%), indicating that the lookahead of MCTS is indispensable for finding profitable trading rules, and neural policy alone are not enough to cope with complex market dynamics. **Exploration is also crucial**: Disabling exploration will reduce the Sharp Ratio to 0.43 (-43%), highlighting the importance of search strategies. **Wasserstein distance is valuable**: Replacing it with a simple MAE (no Wasserstein) will reduce the Sharpe Ratio to 0.22 (-71%), confirming that the distributional alignment is better than the point-wise error minimization. Interestingly, **the influence of memory module and neural network components is moderate** (memory module: Sharpe Ratio 0.15, neural network: Sharpe Ratio 0.45), among which the actual performance of the variant of the neural network is better than expected, indicating that although MCTS provides a core search mechanism, the relative contribution of neural guidance may vary depending on different market regimes.

4.4.4. Wasserstein vs. Alternative Distributional Objectives

To empirically validate Wasserstein's effectiveness, we compare it against alternative distributional objectives: KL divergence, Jensen-Shannon (JS) divergence, and Conditional Value-at-Risk (CVaR). Table 4 reports performance.

Table 4: Performance Comparison Across Distributional Objectives

Objective	AR (%)	SR	MDD (%)	Win Rate
Wasserstein (Full)	**21.2**	**0.76**	**36.1**	**51.8%**
MSE	4.1	0.22	37.3	48.5%
KL Divergence	3.1	0.15	34.8	44.2%
JS Divergence	4.3	0.19	30.9	50.4%
CVaR	9.5	0.43	31.1	51.1%

Finding (RQ1). Wasserstein consistently outperforms other targets in all indicators, especially in tail-risk control (VaR, CVaR) and distribution shape (skewness). This empirical verification supports the theoretical intuition that Wasserstein's transport cost is consistent with the economic utility in trading.

4.4.5. Synthetic Validation

In order to further verify the validity of Wasserstein, we conducted controlled experiments on synthetic return distributions with known properties: Gaussian, skewed, heavy-tail(Student-t) and mixture distributions. For each distribution, we use the Wasserstein target and the MSE objectives to train the EATA variants respectively, and then measure the Earth Mover's Distance (EMD) between the strategy return distribution and the target distribution.

The results show that compared with the strategy of MSE training (mean: 0.032 ± 0.005), the strategy of Wasserstein training achieved a significantly lower EMD score (mean: 0.018 ± 0.003), $p < 0.01$ (pairing t-test). This confirms that the return distribution generated by Wasserstein optimization can better match the shape of the target distribution, especially in the tail area.

4.5. Interpretability and Search Efficiency (RQ2)

4.5.1. Search Efficiency

Hypothesis (RQ2). Neural guidance improves the sample efficiency of symbolic search by biasing the exploration to a promising derivations while retaining the ability to find expressions.

Neural guidance has significantly accelerated the discovery of profit expressions. Figure 6 shows the convergence curve of EATA under the guidance of neural networks. The neural-guided version (Full EATA) achieves a high Sharp Ratio much faster than the pure MCTS version (w/o NN)), confirming that the learned strategy can effectively bias the search to a promising derivation direction.

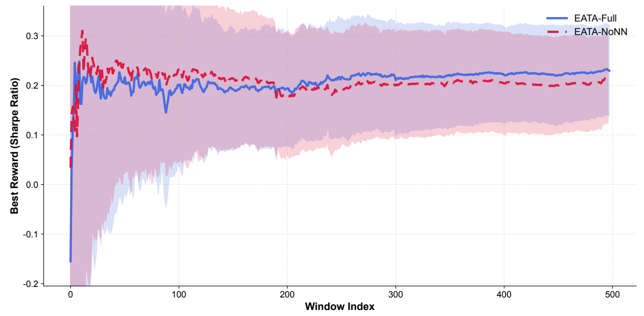

Figure 6: Search Efficiency: Convergence Curves. EATA with neural guidance (Blue) discovers profitable strategies significantly faster than EATA without neural network (Green), demonstrating that neural policy accelerates MCTS search.

4.5.2. Pareto Frontier

Analysis (RQ2/RQ3). The Pareto frontier in Figure 7 shows that EATA achieves superior performance with moderate complexity. EATA generates 6 valid symbol expressions, with an average Sharp Ratio of 0.824 and a complexity of 9-17 AST nodes. In contrast, the average Sharpe ratio of Genetic Programming is only 0.487, and most expressions degenerate to ordinary constants (complexity = 1). In regulated environments, overly complex rules are difficult to audit, so this 1.69× performance advantage at a moderate complexity level is crucial. The Pareto frontier view reveals a trade-off that is often hidden in black-box models: EATA breaks this trade-off by achieving high performance without being too complicated.

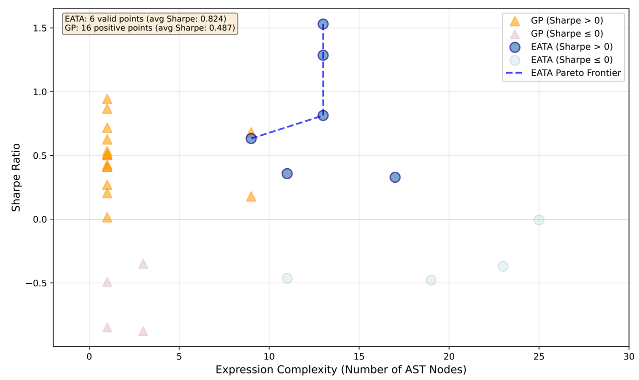

Figure 7: Pareto Frontier of Interpretability vs Performance. EATA (blue circles) achieves superior Sharpe Ratios with moderate expression complexity (9-17 AST nodes) compared to Genetic Programming (orange triangles). The dashed line shows EATA's Pareto frontier, demonstrating the optimal trade-off between interpretability and performance. GP mostly degenerates to trivial constant expressions (complexity=1) with lower average performance.

5. Conclusion

We proposed EATA, a neural-guided symbolic regression framework that explicitly optimizes the trading profitability through Wasserstein distance. Our key insight is that in financial trading, point-wise prediction accuracy (MSE) is not related to economic utility. By formalizing symbolic regression into a distribution matching problem and introducing a dedicated Profit Head into the PVPNet architecture, EATA can find interpretable trading rules with competitive risk-adjusted returns.

We formalize trading as a **distribution matching problem** using Wasserstein distance, which provides a theoretical basis for why this indicator can better capture tail risk and directional asymmetry than MSE or KL divergence. Based on this goal, we have developed a **three-headed PVPNet** (Policy-Value-Profit), and the profit-aware learning of the table obvious formula produces significantly higher risk-adjusted returns than the NoRL variant that only focuses on accuracy (table 3), confirms the value of profit perception at the distribution level. Finally, we conducted **empirical verification** on the S&P 500 universe(N=62 stocks) from 2013 to 2018. In-depth ablation and objective studies confirm the necessity of profit head network and domain-specific grammar, and we provide a The executed assembly line supports the evaluation of the whole stock pool.

There are still some limitations. *computational cost*: Compared with the gradient-based baselines, MCTS will generate a considerable running time per stock, which limits its applicability in high-frequence or large-scale stock pool scenarios. *syntactic design*: The operator set of fixed windows still relies on domain expertise to select windows, and the automatic discovery or adaptation of syntax is left for future work. *single asset focus*: At present, each stock is modeled independently without considering the interactions at the portfolio level or cross-asset structure. *extreme regimes*: Although the sliding window provides a certain robustness for moderate non-stationarity, sudden exogenous shocks may still invalidate the symbol patterns learned in history.

The future work will focus on several complementary directions. First of all, we aim to improve *scalability* by exploring the expression evaluation of parallel MCTS implementation and GPU acceleration. Secondly, we plan to carry out *cross-market validation* in more markets (such as CSI 300, Hang Seng Index) to evaluate the generalization ability outside of U.S.equities. Third, a *portfolio-level extension*, in which the search space explicitly contains cross-asset operators, will be able to find expressions of modeling joint dynamics. Fourth, we will study the *online and continual learning* scheme to update symbol expressions incrementally and reduce the need for complete retraining. Finally, we are interested in integrating the *causal discovery* tool to distinguish between robust structural relationships and spurious correlations.

Explainable transaction systems can enhance regulatory compliance, promote human oversight, and support the domain expert validation. However, the widespread deployment of similar strategies may lead to strategy congestion and reduced effectiveness. Responsible deployment requires monitoring the saturation of the strategy.

Code and data availability. The implementation, experimental configuration and pre-trained checkpoints of EATA can be publicly obtained at https://github.com/JNU-Tangyin/eata. All the data used in this study comes from a public financial database (Yahoo Finance); the processed data set and reproduction script are included in the code base.

Acknowledgment

This work was supported by the National Natural Science Foundation of China (No.62272198) and by the Research and Application of Expressway "1+N" Electromechanical O&M Brain Based on Large Model Agents.

Author Contribution Jingtong Zhang: Conceptualization, Methodology, and Writing Original draft preparation; Xiaoang Yang: Software, Data curation, Visualization; Yin Tang: Supervision, Writing- Reviewing and Editing, Management.

Declaration of generative AI During the preparation of this work the authors used AI for code implementation, polishing, proofreading, and Figure 1 generation from method descriptions. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

[1] T. Fischer, C. Krauss, Deep learning with long short-term memory networks for financial market predictions, European Journal of Operational Research 270 (2) (2018) 654–669.

[2] Z. Zhang, S. Zohren, S. Roberts, Deep learning for portfolio optimization, The Journal of Financial Data Science 2 (4) (2020) 8–20.

[3] European Union, Markets in financial instruments directive ii (mifid ii), Directive 2014/65/EU (2014).

[4] C. Rudin, Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead, Nature Machine Intelligence 1 (5) (2019) 206–215.

[5] W. J. Yeo, W. van der Heever, R. Mao, E. Cambria, R. Satapathy, G. Mengaldo, A comprehensive review on financial explainable ai, arXiv preprint arXiv:2309.11960Submitted on 21 Sep 2023 (2023).

[6] L. Cao, Ai in finance: challenges, techniques, and opportunities, ACM Computing Surveys 55 (3) (2022) 1–38.

[7] R. Cont, Empirical properties of asset returns: stylized facts and statistical issues, Quantitative Finance 1 (2) (2001) 223–236.

[8] J. R. Koza, Genetic programming II: Automatic discovery of reusable programs, MIT Press, 1994.

[9] M. Schmidt, H. Lipson, Distilling free-form natural laws from experimental data, Science 324 (5923) (2009) 81–85.

[10] H. Yang, D. Hao, Z. Wang, Q. Shi, X. Li, Alpha discovery via grammar-guided learning and search, arXiv preprint arXiv:2601.22119 (2026).
URL https://arxiv.org/abs/2601.22119

[11] Y. Xie, T. Qiu, Y. Xiong, X. Huang, X. Gao, C. Chen, An efficient and generalizable symbolic regression method for time series analysis, arXiv preprint arXiv:2409.03986 (2024). doi:10.48550/arXiv.2409.03986.
URL https://arxiv.org/abs/2409.03986

[12] H. Markowitz, Portfolio selection, The Journal of Finance 7 (1) (1952) 77–91. doi:10.1111/j.1540-6261.1952.tb01525.x.
URL https://doi.org/10.1111/j.1540-6261.1952.tb01525.x

[13] W. F. Sharpe, Mutual fund performance, The Journal of Business 39 (S1) (1966) 119. doi:10.1086/294846.
URL https://doi.org/10.1086/294846

[14] A. W. Lo, The statistics of sharpe ratios, Financial Analysts Journal 58 (4) (2002) 36–52. doi:10.2469/faj.v58.n4.2453.
URL https://doi.org/10.2469/faj.v58.n4.2453

[15] O. Ledoit, M. Wolf, Robust performance hypothesis testing with the sharpe ratio, Journal of Empirical Finance 15 (5) (2008) 850–859.

[16] H. White, A reality check for data snooping, Econometrica 68 (5) (2000) 1097–1126. doi:10.1111/1468-0262.00152.
URL https://doi.org/10.1111/1468-0262.00152

[17] P. R. Hansen, A test for superior predictive ability, Journal of Business & Economic Statistics 23 (4) (2005) 365–380. doi:10.1198/073500105000000063.
URL https://doi.org/10.1198/073500105000000063

[18] D. H. Bailey, M. Lopez de Prado, The deflated sharpe ratio: Correcting for selection bias, backtest overfitting and non-normality, SSRN Electronic Journal (2014). doi:10.2139/ssrn.2460551.
URL https://doi.org/10.2139/ssrn.2460551

[19] D. Bailey, J. Borwein, M. López de Prado, Q. J. Zhu, The probability of backtest overfitting, Journal of Computational Finance 20 (4) (2017) 39–69.

[20] C. R. Harvey, Y. Liu, H. Zhu, … and the cross-section of expected returns, The Review of Financial Studies 29 (1) (2016) 5–68. doi:10.1093/rfs/hhv059.
URL https://doi.org/10.1093/rfs/hhv059

[21] D. B. Keim, A. Madhavan, Transactions costs and investment style: an inter-exchange analysis of institutional equity trades, Journal of Financial Economics 46 (3) (1997) 265–292. doi:10.1016/S0304-405X(97)00031-7.
URL https://doi.org/10.1016/S0304-405X(97)00031-7

[22] T. Hendershott, C. M. Jones, A. J. Menkveld, Does algorithmic trading improve liquidity?, The Journal of Finance 66 (1) (2011) 1–33.

[23] C.-H. Park, S. H. Irwin, What do we know about the profitability of technical analysis?, Journal of Economic Surveys 21 (4) (2007) 786–826.

[24] W. Ren, et al., Alpha mining and enhancing via warm start genetic programming for quantitative investment, arXiv preprint arXiv:2412.00896 (2024).

[25] S. Yu, H. Xue, X. Ao, F. Pan, J. He, D. Tu, S. Min, Generating synergistic formulaic alpha collections via reinforcement learning, arXiv preprint arXiv:2306.12964Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (2023).

[26] S. Wang, H. Yuan, L. Zhou, L. M. Ni, H.-Y. Shum, J. Guo, Alpha-gpt: Human-ai interactive alpha mining for quantitative investment, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, Suzhou, China, 2025, arXiv:2308.00016.

[27] H. Shi, W. Song, X. Zhang, J. Shi, C. Luo, X. Ao, et al., Alphaforge: A framework to mine and dynamically combine formulaic alpha factors, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2025, arXiv:2406.18394.

[28] Q. Ding, S. Wu, H. Sun, J. Guo, J. Guo, Hierarchical multi-scale gaussian transformer for stock movement prediction, in: IJCAI, 2020, pp. 4640–4646.

[29] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, in: NeurIPS, 2017, pp. 5998–6008.

[30] Y. Deng, F. Bao, Y. Kong, Z. Ren, Q. Dai, Deep direct reinforcement learning for financial signal representation and trading, IEEE Transactions on Neural Networks and Learning Systems 28 (3) (2016) 653–664.

[31] X.-Y. Liu, H. Yang, Q. Chen, R. Zhang, L. Yang, B. Xiao, C. D. Wang, Finrl: A deep reinforcement learning library for automated stock trading in quantitative finance, arXiv preprint arXiv:2011.09607 (2020).

[32] T. Chen, C. Guestrin, Xgboost: A scalable tree boosting system, in: KDD, 2016, pp. 785–794.

[33] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, Lightgbm: A highly efficient gradient boosting decision tree, in: NeurIPS, 2017, pp. 3146–3154.

[34] H. Deng, Interpreting tree ensembles with intrees, International Journal of Data Science and Analytics 7 (2019) 277–287.

[35] M. T. Ribeiro, S. Singh, C. Guestrin, "why should i trust you?": Explaining the predictions of any classifier, in: KDD, 2016, pp. 1135–1144.

[36] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: NeurIPS, 2017, pp. 4765–4774.

[37] D. Slack, S. Hilgard, E. Jia, S. Singh, H. Lakkaraju, Fooling lime and shap: Adversarial attacks on post hoc explanation methods, in: AIES, 2020, pp. 180–186.

[38] R. Poli, W. B. Langdon, N. F. McPhee, J. R. Koza, A field guide to genetic programming, Lulu.com, 2008.

[39] R. I. McKay, N. X. Hoai, P. A. Whigham, Y. Shan, M. O'Neill, Grammar-based genetic programming: a survey, in: Genetic Programming and Evolvable Machines, Vol. 11, Springer, 2010, pp. 365–396.

[40] T. Stephens, gplearn: Genetic programming in python, https://github.com/trevorstephens/gplearn (2016).

[41] B. Burlacu, G. Kronberger, M. Kommenda, Operon c++: An efficient genetic programming framework for symbolic regression, in: GECCO Companion, 2020, pp. 1562–1570.

[42] S.-H. Chen, Genetic algorithms and genetic programming in computational finance, Springer Science & Business Media (2002).

[43] B. K. Petersen, M. L. Larma, T. N. Mundhenk, C. P. Santiago, S. K. Kim, J. T. Kim, Deep symbolic regression: Recovering mathematical expressions from data via risk-seeking policy gradients, in: International Conference on Learning Representations (ICLR), 2021, pp. 1–12.

[44] L. Biggio, T. Bendinelli, A. Neitz, A. Lucchi, G. Parascandolo, Neural symbolic regression that scales, in: ICML, 2021, pp. 936–945.

[45] P.-A. Kamienny, S. d'Ascoli, G. Lample, F. Charton, End-to-end symbolic regression with transformers, in: NeurIPS, 2022, pp. 10269–10281.

[46] M. Valipour, B. You, M. Panju, A. Ghodsi, Symbolicgpt: A generative transformer model for symbolic regression, in: NeurIPS 2021 AI for Science Workshop, 2021, pp. 1–6.

[47] N. Makke, S. Chawla, Interpretable scientific discovery with symbolic regression: a review, Artificial Intelligence Review 57 (2) (2024) 1–50.

[48] M. J. Kusner, B. Paige, J. M. Hernández-Lobato, Grammar variational autoencoder, in: ICML, 2017, pp. 1945–1954.

[49] K. Ellis, C. Wong, M. Nye, M. Sable-Meyer, L. Morales, L. Hewitt, L. Cary, A. Solar-Lezama, J. B. Tenenbaum, Dreamcoder: Bootstrapping inductive program synthesis with wake-sleep library learning, PLDI (2021) 835–850.

[50] R. Coulom, Efficient selectivity and backup operators in monte-carlo tree search, in: International Conference on Computers and Games, Springer, 2006, pp. 72–83.

[51] L. Kocsis, C. Szepesvári, Bandit based monte-carlo planning, in: ECML, Springer, 2006, pp. 282–293.

[52] D. Silver, A. Huang, C. J. Maddison, A. Guez, L. Sifre, G. Van Den Driessche, J. Schrittwieser, I. Antonoglou, V. Panneershelvam, M. Lanctot, et al., Mastering the game of go with deep neural networks and tree search, Nature 529 (7587) (2016) 484–489.

[53] D. Silver, J. Schrittwieser, K. Simonyan, I. Antonoglou, A. Huang, A. Guez, T. Hubert, L. Baker, M. Lai, A. Bolton, et al., Mastering the game of go without human knowledge, Nature 550 (7676) (2017) 354–359.

[54] D. R. White, S. Yoo, J. Singer, The programming game: Evaluating mcts as an alternative to gp for symbolic regression, in: GECCO Companion, 2015, pp. 1521–1522.

[55] T. Cazenave, Monte-carlo expression discovery, International Journal on Artificial Intelligence Tools 22 (01) (2013) 1250035.

[56] Q. Lu, F. Tao, S. Zhou, et al., Incorporating actor-critic in monte carlo tree search for symbolic regression, Neural Computing and Applications 33 (2021) 8495–8511.

[57] Z. Huang, D. Z. Huang, T. Xiao, D. Ma, Z. Ming, H. Shi, Y. Wen, Improving monte carlo tree search for symbolic regression, in: Advances in Neural Information Processing Systems (NeurIPS), 2025, arXiv:2509.15929.
URL https://arxiv.org/abs/2509.15929

[58] T. Ren, R. Zhou, J. Jiang, J. Liang, Q. Wang, Y. Peng, Riskminer: Discovering formulaic alphas via risk seeking monte carlo tree search, in: ACM International Conference on AI in Finance (ICAIF), 2024. doi:10.1145/3677052.3698613.

[59] F. Sun, Y. Liu, J.-X. Wang, H. Sun, Symbolic physics learner: Discovering governing equations via monte carlo tree search, arXiv preprint arXiv:2205.13134 (2022).

[60] C. Villani, Optimal Transport: Old and New, Vol. 338 of Grundlehren der mathematischen Wissenschaften, Springer, 2009.

[61] G. Peyré, M. Cuturi, Computational optimal transport, Foundations and Trends® in Machine Learning 11 (5-6) (2019) 355–607.
URL https://arxiv.org/abs/1803.00567

[62] S. M. Pesenti, S. Jaimungal, Portfolio optimization within a Wasserstein ball, SIAM Journal on Financial Mathematics 14 (4) (2023) 1175–1214.

[63] V. A. Nguyen, F. Zhang, S. Wang, J. Blanchet, et al., Robustifying conditional portfolio decisions via optimal transport, Operations Research (2025).

[64] M. G. Bellemare, W. Dabney, R. Munos, A distributional perspective on reinforcement learning, arXiv preprint arXiv:1707.06887 (2017). doi:10.48550/arXiv.1707.06887.
URL https://arxiv.org/abs/1707.06887

[65] Y. Chow, A. Tamar, S. Mannor, M. Pavone, Risk-sensitive and robust decision-making: a cvar optimization approach, arXiv preprint arXiv:1506.02188 (2015). doi:10.48550/arXiv.1506.02188.
URL https://arxiv.org/abs/1506.02188

[66] T. Kastner, M. A. Erdogdu, et al., Distributional model equivalence for risk-sensitive reinforcement learning, in: Advances in Neural Information Processing Systems (NeurIPS), 2023.

[67] E. Noorani, C. N. Mavridis, J. S. Baras, Risk-sensitive reinforcement learning with exponential criteria, IEEE Transactions on CyberneticsArXiv:2212.09010 (2025). doi:10.1109/TCYB.2025.3575240.

[68] C. Ma, F. Gao, L. Ji, C. Zhang, L. Long, J. Zhang, Towards personalized risk-sensitive decision-making: A novel risk preference adaptive distributional reinforcement learning algorithm for stock trading, Applied Soft Computing 156 (2025) 114269. doi:10.1016/j.asoc.2025.114269.

[69] J. Zhang, Tail-safe hedging: Explainable risk-sensitive reinforcement learning with a white-box cbf–qp safety layer in arbitrage-free markets, arXiv preprint arXiv:2510.04555 (2025).
URL https://arxiv.org/abs/2510.04555

[70] B. Goodman, S. Flaxman, European union regulations on algorithmic decision-making and a "right to explanation", AI Magazine 38 (3) (2017) 50–57.

[71] S. Jain, B. C. Wallace, Attention is not explanation, in: NAACL-HLT, 2019, pp. 3543–3556.

[72] S. Serrano, N. A. Smith, Is attention interpretable?, in: ACL, 2019, pp. 2931–2951.

[73] S. Wiegreffe, Y. Pinter, Attention is not not explanation, in: EMNLP-IJCNLP, 2019, pp. 11–20.

[74] D. Alvarez-Melis, T. S. Jaakkola, Towards robust interpretability with self-explaining neural networks, in: NeurIPS, 2018, pp. 7775–7784.

[75] G. Appel, Technical analysis: Power tools for active investors, FT Press, 2005.

[76] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Computation 9 (8) (1997) 1735–1780.

[77] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, arXiv preprint arXiv:1707.06347 (2017) 1–12.

Appendix A. Hyperparameters

Table A.5 lists the hyperparameters used for baseline methods. We used standard parameter settings from the literature and FinRL default configurations.. For traditional and statistical baselines, we used standard parameter settings from the literature.

Table A.6 lists the default hyperparameters used in EATA experiments.

Table A.5: Baseline Hyperparameters

Method	Hyperparameter	Value
LightGBM	learning_rate	0.1
	num_leaves	31
	max_depth	-1
LSTM	hidden_size	128
	num_layers	2
	dropout	0.2
Transformer	d_model	64
	nhead	4
	num_layers	2
PPO	learning_rate	3e-4
	gamma	0.99
	clip_range	0.2
A2C	learning_rate	3e-4
	gamma	0.99
	entropy_coef	0.01
TD3	learning_rate	3e-4
	gamma	0.99
	tau	0.005
DDPG	learning_rate	3e-4
	gamma	0.99
	tau	0.005
ARIMA	order	(5,1,0)
	seasonal_order	(0,0,0,0)
MACD	fast_period	12
	slow_period	26
	signal_period	9
Genetic Programming	population_size	100
	generations	50

Table A.6: EATA Hyperparameters

Category	Parameter	Value
MCTS	Episodes per iteration (K)	100
	Max expression length ($L^{\max}$)	50
	UCB exploration constant (C)	$1/\sqrt{2} \approx 0.707$
Neural Network	Hidden dimension (d_h)	16
	Learning rate	10^{-5}
	Batch size	32
Loss Weights	λ^{π} (Policy)	1.0
	λ^{V} (Value)	1.0
	λ^{P} (Profit)	1.0
Training	Sliding window size (T^{in})	100 days
	Lookahead (H)	20 days
	Buffer capacity ($B^{\max}$)	10,240
Grammar	Module max length (L^{mod})	10
	Module library size ($M^{\max}$)	100
	Quantile thresholds (Q_{25}, Q_{75})	0.25, 0.75

Appendix B. Wasserstein Distance

Proposition 1. *Wasserstein distance is superior to MSE and KL divergence for financial returns due to three properties: metric structure, tail sensitivity, and geometric interpretation.*

Proof. Property 1 (Metric Structure). The 1-Wasserstein distance satisfies:

1. Non-negativity: $\mathcal{W}_1(P,Q) \geq 0$
2. Identity: $\mathcal{W}_1(P,Q) = 0 \iff P = Q$
3. Symmetry: $\mathcal{W}_1(P,Q) = \mathcal{W}_1(Q,P)$
4. Triangle inequality: $\mathcal{W}_1(P,R) \leq \mathcal{W}_1(P,Q) + \mathcal{W}_1(Q,R)$

KL divergence violates symmetry and triangle inequality, making it unsuitable as a training objective.

Property 2 (Tail Sensitivity). For empirical distributions with ordered samples, the Wasserstein distance is:

$$\mathcal{W}_1 = \frac{1}{N}\sum_{i=1}^{N} |x_{(i)} - y_{(i)}| \tag{B.1}$$

This equally weights all quantiles, including extremes. MSE, by contrast, is dominated by the mean: $\text{MSE} = \frac{1}{N}\sum(x_i - y_i)^2$.

Property 3 (Geometric Interpretation). Wasserstein distance measures the minimal cost of transporting probability mass from P to Q, respecting the metric structure of the underlying space. This is crucial for returns where the magnitude (not just probability) of extreme events matters. □

□

Appendix C. Theoretical Properties of the Wasserstein Training Objective

We analyze three theoretical properties of the 1-Wasserstein reward $r^{\text{rl}} = 1/(1 + \mathcal{W}_1)$: gradient characterization, convergence guarantees, and CVaR control.

Appendix C.1. Gradient Characterization in One Dimension

For one-dimensional distributions, the 1-Wasserstein distance admits a closed-form representation via CDFs [60]:

$$\mathcal{W}_1(P, Q) = \int_{-\infty}^{\infty} |F_P(x) - F_Q(x)|\, dx \tag{C.1}$$

$$= \int_0^1 |F_P^{-1}(u) - F_Q^{-1}(u)|\, du \tag{C.2}$$

where F_P^{-1}, F_Q^{-1} are quantile functions. The functional derivative with respect to the predicted CDF F_θ is:

$$\frac{\delta \mathcal{W}_1}{\delta F_\theta}(x) = \text{sgn}(F_\theta(x) - F^*(x)), \tag{C.3}$$

yielding a clear training signal: overestimating the true CDF at x pushes $F_\theta(x)$ downward, and vice versa. For the reward $r^{\text{rl}} = 1/(1 + \mathcal{W}_1)$, the gradient:

$$\frac{\partial r^{\text{rl}}}{\partial \theta} = -\frac{1}{(1 + \mathcal{W}_1)^2} \cdot \frac{\partial \mathcal{W}_1}{\partial \theta} \tag{C.4}$$

preserves direction with adaptive scaling.

Appendix C.2. Convergence and Sample Complexity

$\mathcal{W}_1$ metrizes weak convergence on spaces with bounded first moment [60]. For empirical distributions with n, m i.i.d. samples,

$$\mathbb{E}\big[|\mathcal{W}_1(\hat{P}_n, \hat{Q}_m) - \mathcal{W}_1(P, Q)|\big] = O(n^{-1/2} + m^{-1/2}), \tag{C.5}$$

under finite second moment. This contrasts favorably with KL divergence (unbounded under mismatch) and MSE (degraded under heavy tails [7]).

Appendix C.3. Relationship with Conditional Value-at-Risk (CVaR)

Proposition 2 (CVaR Bound via Wasserstein Distance)**.** *Let P and Q be probability distributions on $\mathbb{R}$ with finite first moments. Then for any $\alpha \in (0, 1)$:*

$$|CVaR_\alpha(P) - CVaR_\alpha(Q)| \leq \frac{1}{1 - \alpha} \cdot \mathcal{W}_1(P, Q). \tag{C.6}$$

Proof. From the quantile representation and the definition:

$$\text{CVaR}_\alpha(P) = \frac{1}{1 - \alpha} \int_\alpha^1 F_P^{-1}(u)\, du, \tag{C.7}$$

we have:

$$\begin{aligned} |\text{CVaR}_\alpha(P) - \text{CVaR}_\alpha(Q)| &\leq \frac{1}{1 - \alpha} \int_\alpha^1 |F_P^{-1}(u) - F_Q^{-1}(u)|\, du \\ &\leq \frac{1}{1 - \alpha} \int_0^1 |F_P^{-1}(u) - F_Q^{-1}(u)|\, du \\ &= \frac{1}{1 - \alpha} \cdot \mathcal{W}_1(P, Q). \quad \square \end{aligned} \tag{C.8}$$

□

For $\alpha = 0.95$, the bound is $20 \cdot \mathcal{W}_1$, so $\mathcal{W}_1 = 0.01$ guarantees $\text{CVaR}_{0.95}$ agreement within 0.2 units. This provides α-agnostic tail regularization without explicit CVaR tuning [65].

Appendix C.4. Comparison with Alternative Divergences

Table C.7: Theoretical Comparison of Distributional Divergences

Property	$\mathcal{W}_1$	**KL**	**JS**
Symmetric	✓	×	✓
Triangle inequality (metric)	✓	×	×
Well-defined for disjoint supports	✓	×	×
Sensitive to geometry of $\mathbb{R}$	✓	×	×
Bounded CVaR gap	✓	×	×
Parametric sample complexity (1D)	$O(n^{-1/2})$	Unbounded	Unbounded

Appendix D. Additional Discovered Expressions

Volatility-Adjusted Momentum.

$$e = \frac{\text{mom}(x_{\text{close}}, 10)}{\text{vol}(x_{\text{close}}, 20)} \times \text{ite}\big(\text{ma}(x_{\text{close}}, 5) > \text{ma}(x_{\text{close}}, 20),\ 1,\ 0\big) \tag{D.1}$$

Interpretation: Momentum normalized by volatility, filtered by trend confirmation (golden cross).

Mean Reversion with Volume.

$$e = \big(x_{\text{close}} - \text{ma}(x_{\text{close}}, 20)\big) \times \frac{x_{\text{volume}}}{\text{ma}(x_{\text{volume}}, 20)} \tag{D.2}$$

Interpretation: Price deviation scaled by relative volume (higher conviction on high volume).